

Inclusion-Exclusion Principle:-

The inclusion-exclusion principle is concerned with the number of elements in the set operations. Since in a set an element is to be counted once the principle states that while counting the elements in a situation if some elements are not counted already they are to be included and if some elements are already counted they are to be excluded.

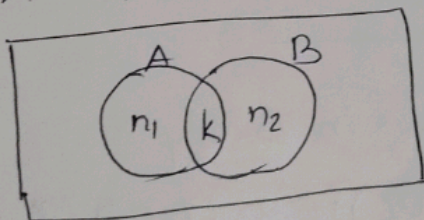
For example if we want the number of elements in $A \cup B$ denoted by $n(A \cup B)$ we add $n(A)$ and $n(B)$ and subtract $n(A \cap B)$ i.e. in our counting we include $n(A)$ and $n(B)$ and exclude $n(A \cap B)$ to avoid double counting because they are counted in $n(A)$ as well as in $n(B)$. This is inclusion-exclusion principle.

For two sets

$$(i) n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

if A and B are disjoint sets then $n(A \cap B) = 0$

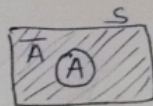
$$(ii) \therefore n(A \cup B) = n(A) + n(B)$$



For three sets

$$(iii) n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$$

$$(iv) n(\bar{A}) = n(S) - n(A)$$

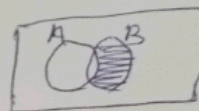
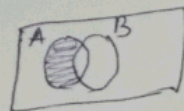


$$(v) n(A - B) = n(A \cap \bar{B}) = n(A) - n(A \cap B)$$

The no. of elements in A but not in B

$$(vi) n(B - A) = n(B \cap \bar{A}) = n(B) - n(A \cap B)$$

The no. of elements in B but not in A .

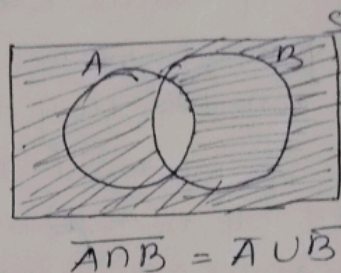
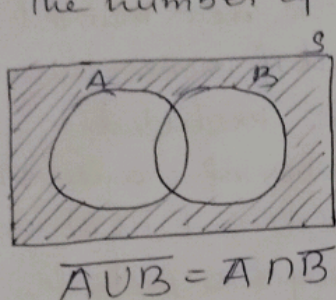


$$(VI) \quad n(A \Delta B) = n(A - B) + n(B - A) \\ = n(A \cap \bar{B}) + n(B \cap \bar{A})$$

The number of elements in only A or in only B is equal to the sum of (v) & (vi)

$$(VII) \quad n(\overline{A \cup B}) = n(\bar{A} \cap \bar{B}) = n(S) - n(A \cup B)$$

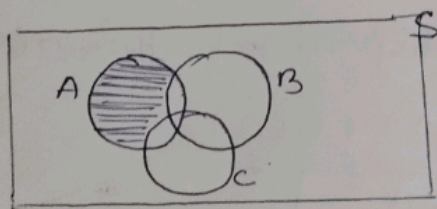
The number of elements neither in A nor in B.



$$(IX) \quad n(\overline{A \cap B}) = n(\bar{A} \cup \bar{B}) = n(S) - n(A \cap B)$$

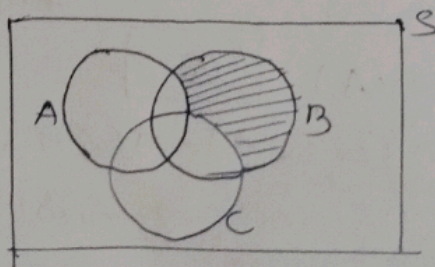
(X) The number of elements in A only

$$n(A - B - C) = n[A \cap (\overline{B \cup C})] = n[A \cap \bar{B} \cap \bar{C}] \\ = n(A) - n(A \cap B) - n(A \cap C) + n(A \cap B \cap C)$$

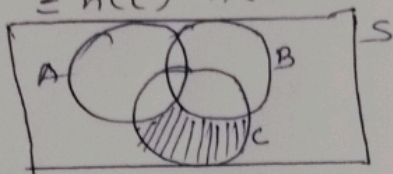


(XI) The number of elements in B only

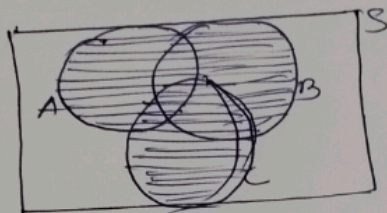
$$n(B - A - C) = n[B \cap (\overline{A \cup C})] = n(B \cap \bar{A} \cap \bar{C}) \\ = n(B) - n(B \cap A) - n(B \cap C) + n(A \cap B \cap C)$$



- (XII) The number of elements in C only
 $n(C - A - B) = n[C \cap (A \cup B)^c] = n[C \cap \bar{A} \cap \bar{B}]$
 $= n(C) - n(C \cap A) - n(C \cap B) + n(A \cap B \cap C)$



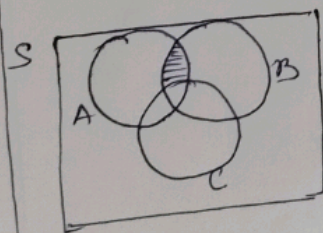
- (XIII) The number of elements in A or B or C
 $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$



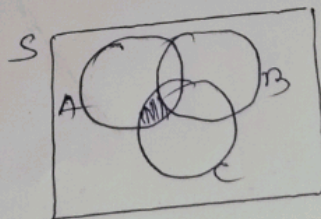
- (XIV) The number of elements in A and B only but not in C
 $n(A \cap B \cap \bar{C}) = n(A \cap B) - n(A \cap B \cap C)$

- (XV) The number of elements in A and C only but not in B
 $n(A \cap C \cap \bar{B}) = n(A \cap C) - n(A \cap B \cap C)$

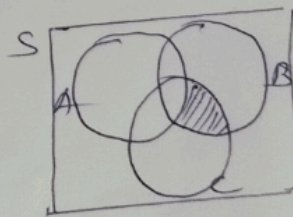
- (XVI) The number of elements in B and C only but not in A
 $n(B \cap C \cap \bar{A}) = n(B \cap C) - n(B \cap C \cap A)$



(XIV)



(XV)



(XVI)

- (XVII) The number of elements neither in A nor in B nor in C
 $n(\overline{A \cup B \cup C}) = n(\bar{A} \cap \bar{B} \cap \bar{C}) = n(S) - n(A \cup B \cup C)$